

Part 2 of 2

Host Software for RFID Based Attendance Management System



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In the first part, we covered designing the user interface (UI) program, but that is not complete unless we include the service layer project and enable client-server interaction. This part talks about these two aspects, along with hardware interfacing and testing of the project on a Windows PC.

Designing the service layer

To develop this application, we use the concept of service oriented architecture (SOA), which makes our server-side logic independent of client logic.

WCF is a framework that supports SOA. WCF publishes the CLR object as a service, which can be used among known and unknown runtimes, irrespective of their locations.

We use simple object access protocol (SOAP) based service. SOAP is mainly an XML document that provides an envelope for our messages. It is transport- and platform-independent. A WCF service has web service definition language (WSDL), which is again an XML document that describes how to communicate with the service. Specifics of how to communicate with a service is given by its service endpoint. There are three elements that constitute a service endpoint: address, binding and contract. A contract is platform-neutral and a standard way of describing what a service does.

In this application, we mainly use ServiceContract and OperationContract.

1. Create a class library AttendanceRegistrationServiceContractsLib (refer Fig. 21) in the solution. Then, create an interface IAttendanceRegisterService inside the class library and make it a service contract by adding ServiceContract attribute to make it identifiable to the client-side as a service. Add abstract methods whose functionalities are to be provided by the server-side to client-side as OperationContract attribute.

Add DataTransferObjectLib reference to this project. WCF contracts are present inside System.ServiceModel namespace, which can be added from .Net tab of Add Reference

methods using that data-access layer. Add reference of libraries DataTransferObjectLib, AttendanceRegistrationServiceContractsLib, AttendanceDataAccessLib, AttendanceDataRepository and System.ServiceModel namespace from .Net tab of Add References dialog box.

3. Finally, create a console application AttendanceRegisterServer and inside it create a ServiceHost, which is responsible for creating instances of the service. ServiceHost class is used to configure and expose a service for use by client applications when we are not using web servers, Internet information services (IIS) or Windows activation services (WAS) to expose a service. Both IIS and WAS interact with ServiceHost object on our behalf. System.ServiceModel.ServiceHost namespace holds this object, which is created while self-hosting the WCF service.

Now, create the service endpoint. Here, we use BasicHttpBinding as the communication stack (protocol) to communicate with other client applications. BasicHttpBinding is a binding that WCF service can use to configure and expose endpoints. For the address of our service, use a URL based on the port number of our system.

Then, describe the WSDL that contains the metadata of all our endpoints configured in that system. This WSDL will be accessed by other client applications when these need to access certain server functionalities.

Now, set the mode of our service host to Open.

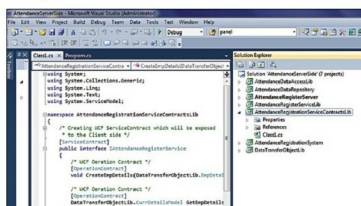


Fig. 21: AttendanceRegistrationServiceContractsLib

es dialog box. At the beginning of file Class1.cs of AttendanceRegistrationServiceContractsLib project, add the line "using System.ServiceModel;"

2. Create another class library AttendanceRegisterServiceLib and rename the class inside it to AttendanceRegisterService, which is inherited from IAttendanceRegisterService interface of AttendanceRegistrationServiceContractsLib namespace. Inside the class, create an instance of the data-access layer inside its constructor and define all interface